RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Thursday, 24 September 2026, 10:58 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:16 AM
Time taken	18 mins 5 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int max(int a, int b) {
    return (a > b) ? a : b;
}
int main(){
    char s1[1000], s2[1000];
    if(scanf("%s",s1) !=1)return 0;
    if(scanf("%s",s2)!=1)return 0;
    int m = strlen(s1);
    int n = strlen(s2);
    int dp[m+1][n+1];
    for(int i = 0;i<=m;i++){
        for(int j=0; j<=n; j++){
            if(i ==0 || j ==0){
                dp[i][j]=0;
            }else if (s1[i-1]==s2[j-1]){
                dp[i][j]=1+dp[i-1][j-1];

            }else{
                dp[i][j] = max(dp[i-1][j], dp[i][j-1]);
            }
        }
    }
    printf("%d\n", dp[m][n]);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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